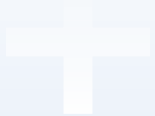


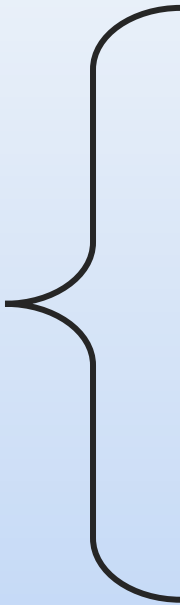
Blog - 01

Google Gemini: Backend Workflow From Prompts to Execution

A Deep Dive into the AI Engine Driving Gemini

Agenda



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- **About Gemini**
 - **Purpose** ("why")
 - **Scope** ("where")
 - **Timeline** ("when")
 - **Stakeholders / Team Roles** ("who")
 - **Backend Workflow** ("how")
 - **Impact Analysis**
 - **Real world application**

Summary

What is Gemini?



Gemini (also known as Google Gemini) is a generative artificial intelligence chatbot and virtual assistant developed by Google.

It is a multimodal AI Foundation capable of understanding and generating content across multiple formats, including text, images, audio, video, and code.

Unlike traditional AI systems that specialize in only one type of data, Gemini is designed to process different forms of information within a simplified architecture.

Google uses to teach Gemini how to process and generate human-like responses through a multi-layered process



1.

Purpose ("why")

Gemini was developed to solve four main limitations of earlier "GPT" models:

- **Native Multimodality:** Instead of bolting on image-to-text translators, Gemini processes text, code, audio, video, and images directly in a single, unified neural network.
- **Massive Context Window:** Early models forgot things quickly. Gemini models can hold millions of tokens in their working memory (like thousands of pages of text, hours of video, or entire codebases) in a single prompt.
- **Google Ecosystem:** It was built specifically to power Google's massive infrastructure, running everywhere from Workspace apps(docs, maps , Gmail) to on-device Android phones and may runs without internet access.
- **Hardware Efficiency:** It is optimized to run on Google's custom AI chips (TPUs) for better speed and scalability compared to relying solely on standard GPUs.

It aims to make human-computer interaction more natural and efficient.

2.

Scope ("where")

Gemini was developed by **Google**, specifically by its AI research division **Google DeepMind** (an artificial intelligence research laboratory within Google's parent company), which was formed when Google's DeepMind and Google Brain teams merged in 2023.

Gemini has a very broad scope because it is designed to help in many areas.

1. **Education** - Explains concepts, Summarizes books and articles.
2. **Programming** -Writes code , Finds bugs, Explains algorithms.
3. **Business**- Writes emails, Analyzes data, Automates routine work.
- 4.**Healthcare** -Summarizes medical literature, Assists researchers.
5. **Creative Work**-Writing stories and poems, Brainstorming ideas, Creating marketing content.
- 6.**Google Products**- Gmail ,Google Docs, Google Sheets, Google Maps , Google Search



3.

Timeline ("when")

2023 – Gemini announced

- On **December 6, 2023**, Google officially announced the Gemini family of AI models. Google introduced three versions:
 - Gemini Ultra** – the most powerful model.
 - Gemini Pro** – for a wide range of tasks.
 - Gemini Nano** – designed to run efficiently on devices like smartphones

2024 – Public rollout

- In **February 2024**, Google renamed its AI chatbot from **Bard** to **Gemini**.

The Gemini app became available on Android, and users could also access it on the web.

Throughout 2024, Google integrated Gemini into products like Gmail, Docs, Sheets, and other Workspace applications.

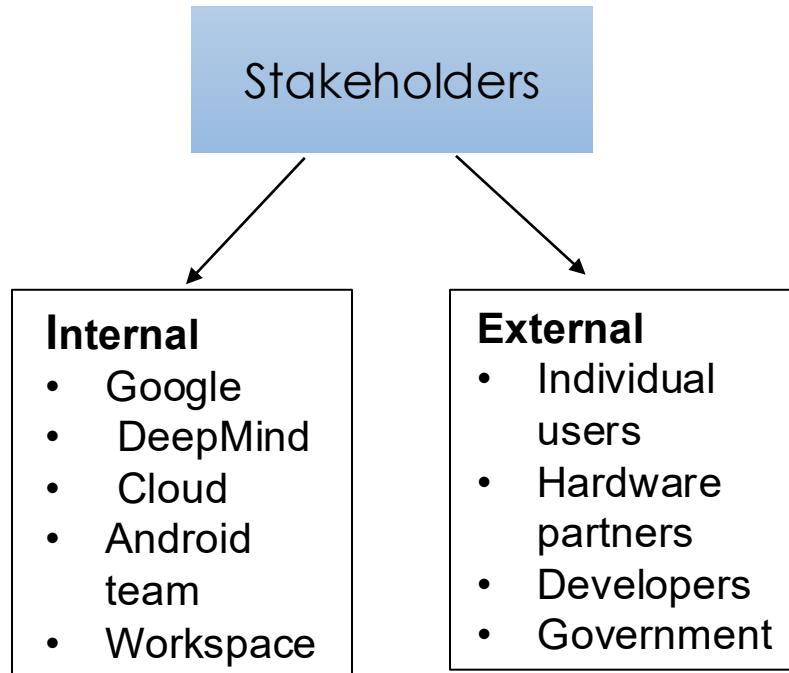
2025 and beyond

- Google continued releasing newer Gemini models with improved reasoning, coding, multimodal capabilities, and larger context windows.
- Gemini became a core AI technology across many Google products and services.

4.

Stakeholders / Team Roles ("who")

How did the idea come about? The idea came from the rapid progress of AI models like GPT. Google had already built many important AI technologies, including the Transformer architecture introduced in the 2017 paper *Attention Is All You Need*. That architecture became the foundation for many modern language models.



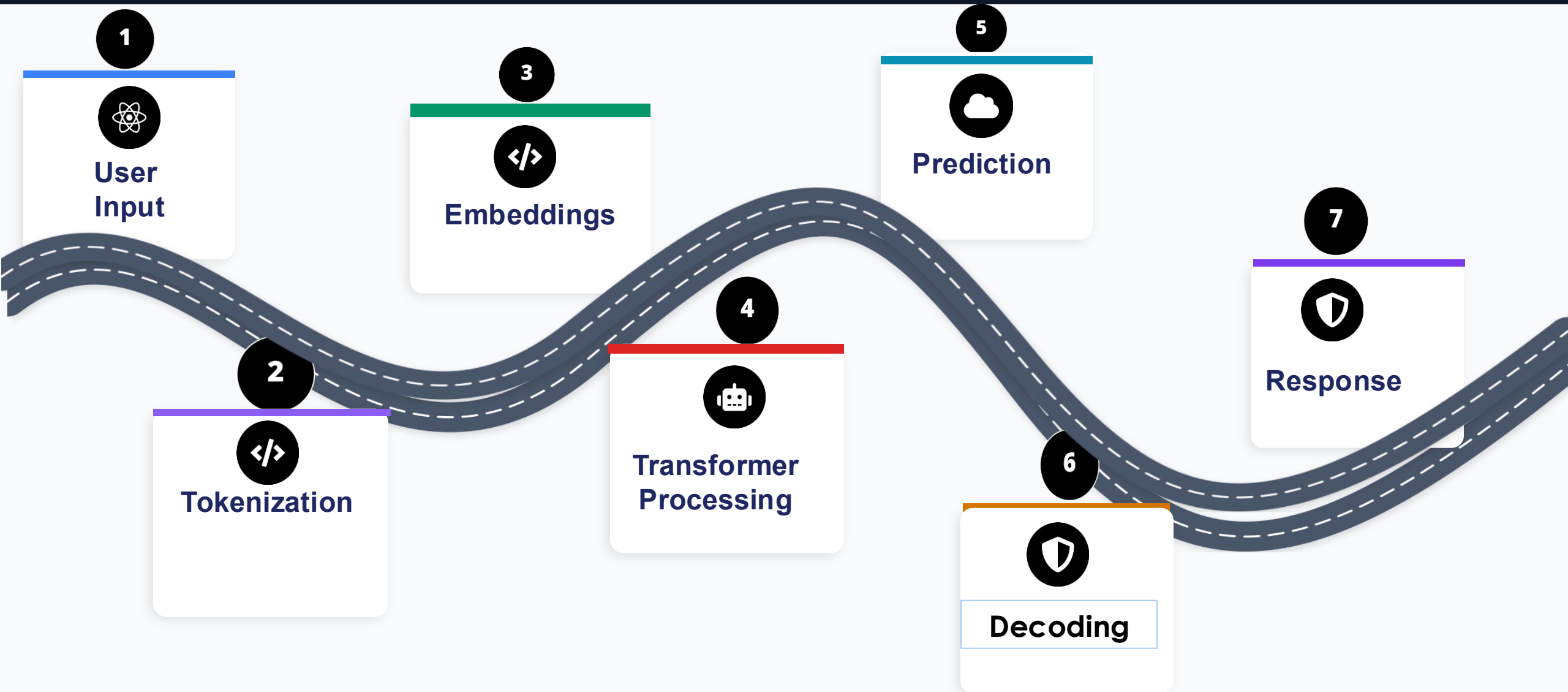
Indian Contribution

- Google offices in Bengaluru and Hyderabad contributed through engineering, AI research, cloud infrastructure, testing, and support for Indian languages.
- Many leading AI researchers at Google are also of Indian origin, although Gemini is the result of a global team rather than the work of any one nationality.

Key leaders

- **Demis Hassabis** – Leads Google DeepMind and has been a principal leader behind Gemini.
- **Jeff Dean** – most influential AI and systems researchers.
- **Sundar Pichai** – Provides overall strategic direction and announced Gemini publicly.

Backend Workflow ("how")



Step 1: Input Reception

The process begins when a user submits a query.

Example: "What is machine learning?"

The input may come from:

- Web browser
- Mobile application
- Voice assistant
- Google products

The system receives the request and forwards it to Gemini's inference servers

Step 2 :Tokenization

Computers do not understand language directly. The text must first be broken into smaller units called tokens.

Example:

Input: "Artificial Intelligence is amazing."

Possible tokens: ["Artificial", "Intelligence", "is", "amazing", "."]

Each token receives a unique numerical identifier.

Example: Artificial → 2341

Intelligence → 9187

is → 55

amazing → 1024

This conversion allows the model to process language mathematically.

Step 3: Embedding Generation

After tokenization, tokens are converted into dense numerical vectors called embeddings.

Example: Token: "cat"

Embedding: [0.32, -0.12, 0.84, 0.67, ...]

These vectors represent semantic meaning. Words with similar meanings tend to have similar embeddings. Example: Cat, Kitten, Feline

Their vector representations appear close together in mathematical space. Embeddings help Gemini understand relationships between concepts.

Step 4: Transformer Processing

The transformer is the core engine of Gemini. It analyzes relationships among tokens and extracts meaning.

A. Attention Mechanism Attention helps the model focus on relevant words. Example: "The animal didn't cross the road because it was tired." The model understands that "it" refers to "animal."

B. Context Understanding The transformer examines all tokens simultaneously. Unlike older sequential models, transformers process entire sentences in parallel

C. Pattern Recognition During training, Gemini learns billions of language patterns. For example: Input: "Paris is the capital of" Prediction: "France" This ability comes from recognizing statistical patterns learned during training.

Step 5: Context Window

The context window acts as Gemini's short-term memory.

It stores: Current prompt Previous messages Relevant conversation history
Example:

User: "Who invented the telephone?"

Gemini: "Alexander Graham Bell."

User: "When was he born?"

Gemini understands that "he" refers to Alexander Graham Bell because earlier conversation remains within the context window. A larger context window enables Gemini to analyze lengthy documents and maintain coherent conversations.

Step 6: Prediction Engine

After processing context,

Gemini predicts the most probable next token.

Example:

Input: "Machine learning is a branch of"

Possible predictions:

Computer science → 75% Mathematics
→ 15%

Engineering → 7%

Physics → 3%

The model selects from these probability distributions. This process occurs repeatedly until the response is complete.

Step 7: Decoding

The model does not always choose the highest-probability token. Instead, decoding strategies improve response quality.

A. Temperature Temperature controls randomness. Low Temperature (0.2) More predictable More factual High Temperature (1.0) More creative More diverse responses

B. Top-K Sampling The model considers only the top K most probable tokens.

Example: K = 5 Only the five highest-probability words are eligible for selection.

C. Top-P Sampling Instead of a fixed number of words, Top-P selects tokens whose cumulative probability exceeds a threshold. Example: P = 0.9 The model chooses from tokens representing 90% of total probability. These methods balance creativity and accuracy.

Step 8: Response Generation

Once tokens are selected, they are combined into sentences.

Example:

Predicted tokens:

["Machine", "learning", "is", "a", "branch", "of", "AI"]

Generated output: "Machine learning is a branch of AI."

The response is built token-by-token until completion. Additional safety filters may evaluate the response before release. The final response is transmitted back to the user interface- Gemini application .

Multimodal Processing Capabilities



Text Understanding and Generation

Transformer-based architecture

Multi-layer attention for meaning, syntax, and context

Dynamic tokenization for multilingual and technical text

Iterative generation with coherence refinement



Image Analysis and Visual Reasoning

Convolutional + transformer vision system

Multi-resolution detail and global context

Cross-modal alignment for descriptions and visual Q&A

Spatial relationships and abstract visual concepts



Code Generation and Programming Logic

Syntax, semantics, and execution flow

Code generation, debugging, refactoring, optimization

Multi-language support and translation



Audio Processing and Speech Recognition

Signal processing and neural audio codecs

Spectral representations of time and frequency

Noise reduction and speaker separation



Video Content Analysis

3D convolution for motion and temporal patterns

Object tracking, scene segmentation, activity recognition

Impact Analysis

Gemini has the potential to improve learning, productivity, and innovation across many industries.

However, it should be used responsibly, with human oversight for important decisions, to minimize risks and ensure accurate, ethical use

Real world application

- **Education** – Explains concepts, solves questions, summarizes notes, and helps with homework.
- **Programming** – Writes code, fixes bugs, and explains programming concepts.
- **Business** – Drafts emails, creates reports, analyzes data, and prepares presentations.
- **Healthcare** – Assists with medical research, summarizes clinical information, and supports documentation (under professional supervision).
- **Customer Support** – Powers AI chatbots to answer customer queries 24/7.
- **Content Creation** – Helps write articles, blogs, social media posts, and marketing content.
- **Google Workspace** – Assists users in Gmail, Docs, Sheets, and Slides.
- **Android Devices** – Acts as an AI assistant for voice commands, searches, and productivity tasks.
- **Research** – Summarizes research papers and analyzes large amounts of information.
- **Translation** – Translates text and supports communication across multiple languages.

Summary

Google Gemini represents a significant leap forward in AI technology, built on sophisticated architecture that handles multiple types of data seamlessly. Its advanced training methods and multimodal processing allow it to understand text, images, and code with remarkable accuracy. The system's memory management keeps conversations flowing naturally while maintaining context across complex interactions.

As AI technology continues to evolve, Google Gemini will play a significant role in shaping the future of human-computer interaction and intelligent digital assistance.



Conclusion



Multimodal Architecture

Handles text, images, and code seamlessly.



Advanced Training

Understands multiple data types with remarkable accuracy.



Memory Management

Keeps conversations flowing naturally and maintains context.



Safety and Efficiency

Built-in alignment framework and performance optimizations.



Google Gemini is just getting started.

Thank you

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